Is Differential Radial Flow aka $v_0(p_T)$ Collective? Role Of Multiplicity Fluctuations And Non-Flow Bias

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Introduction

- Radial flow quantified as the dynamical correlation in transverse momenta, p_T, among outgoing hadrons in a Heavy-Ion event.
- Sensitive to initial state fluctuations, QGP bulk viscosity and equation of state. Usually accessed via $[p_T]$ cumulants, $[p_T]$: p_T per particle in an event.
- Nature of Radial flow fluctuations: Property of single-particle (collective nature) or multi-particle distribution? Still unknown.
  - Cannot be resolved using integrated objects like $[p_T]$ cumulants → Requires a differential probe for radial flow fluctuations.

The Differential Observable

- Integrated radial flow defined as square root of normalized variance of $[p_T]$ fluctuations.

$$\frac{\sqrt{(\delta[p_T])^2}}{\langle[p_T]\rangle} = v_0$$

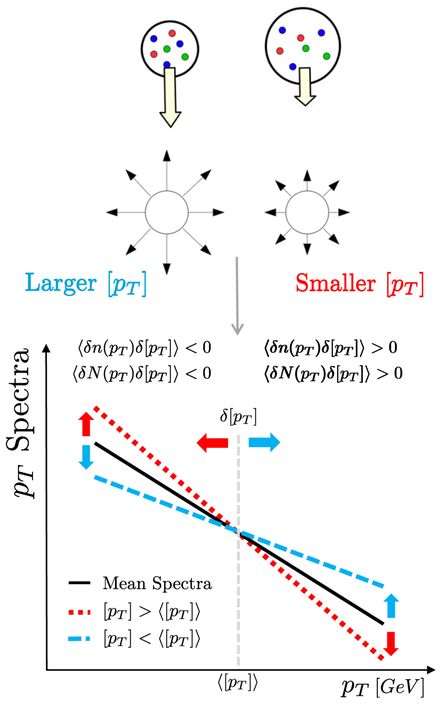
- Two possible formulations for differential radial flow based on full, N(p_T) or normalized, n(p_T) spectra.

$$\frac{\langle\delta N(p_T)\delta[p_T]\rangle}{\langle N(p_T)\rangle\langle[p_T]\rangle} = v_0'(p_T)v_0$$

$$\frac{\langle\delta n(p_T)\delta[p_T]\rangle}{\langle n(p_T)\rangle\langle[p_T]\rangle} = v_0(p_T)v_0$$

⟨ ⟩ denotes event ensemble average

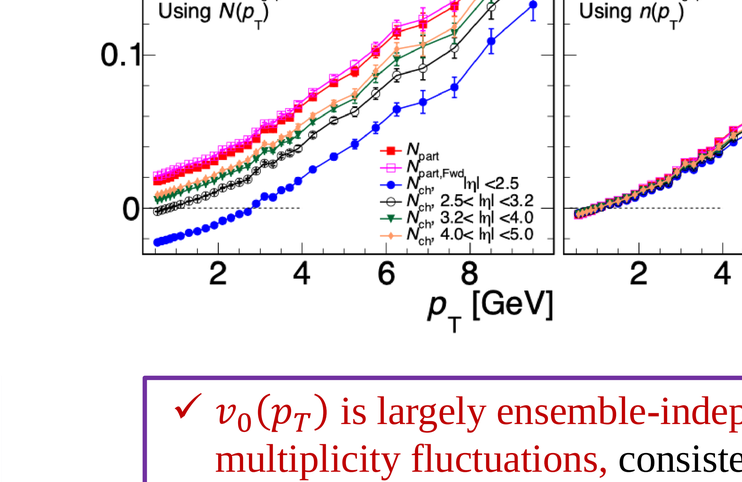
- Probes the correlated fluctuation between local spectra and $[p_T]$.



Current Physics Interpretation

Ⓐ Factorization Behavior: Invariant to changes in p_T^{ref} → Hallmark of Collective Origin for radial flow.

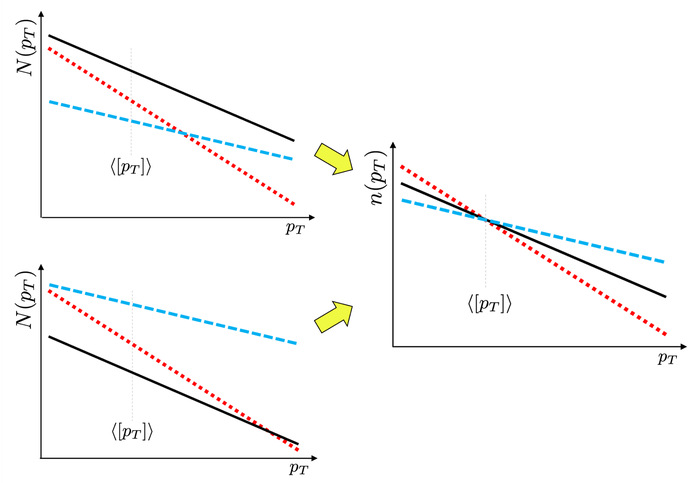
Ⓑ Split for p_T > 4 GeV → Transition from Flow dominated to Jet Quenching dominated regime.



ATLAS, arXiv:2503.24125

What More Can be Probed?

- $v_0'(p_T)$ sensitive to multiplicity fluctuations for given event classifier.
- New lever arm to quantify fluctuations in particle production process with changing rapidity.



$$v_0'(p_T) - v_0(p_T) \approx \frac{\langle\delta N\delta[p_T]\rangle}{\langle N\rangle\sigma_{p_T}}$$

Results

We investigate the non-flow bias and the potential to constrain multiplicity fluctuations within the HIJING Model with a p_T-hard scale of 2 GeV in the default configuration.

Impact Of Non-Flow Bias

- ✓ Factorization Behavior Breaking: $v_0(p_T)$ in HIJING sensitive to p_T range of refence particles, different zero-crossing values.
- ✓ $v_0(p_T)$ rises nearly linearly at high p_T , consistent with expectations from an exponential spectrum.

Impact Of Multiplicity Fluctuations

- ✓ $v_0(p_T)$ is largely ensemble-independent, while $v_0'(p_T)$ is sensitive to global multiplicity fluctuations, consistent with expectations.
- ✓ The difference, quantified by $\langle\delta N\delta[p_T]\rangle$, is constant in p_T.
- ✓ Minor differences persist across N_{ch} and N_{part} based measurements at low p_T after applying correction.

Conclusion And Results

- Both the splitting behavior and the linear increase for peripheral centrality at large p_T for $v_0(p_T)$ in data can be attributed to non-flow contamination.
- The difference between the two formulations of differential radial flow ($v_0'(p_T)$ - $v_0(p_T)$) can be used to isolate and understand multiplicity fluctuations.

References

1) ATLAS, arXiv:2503.24125
2) B. Schenke, C. Shen and D. Teaney, PRC 102 (2020) 034905
3) T. Parida, R. Samanta and J.Y. Ollitrault, PLB 857 (2024) 138985